KRISHNANUNNI C G

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RESEARCH INTERESTS

My long-term goal is to pursue independent research and teaching at the intersection of computational mathematics and machine learning. My interests include neural architecture design, inverse problems, theoretical foundations of machine learning, and the development of mathematically principled algorithms for scientific machine learning.

CURRENT POSITION

Postdoctoral Researcher **August 2026 – Present**
Institute for Mathematics, Heidelberg University, Germany

ACADEMIC BACKGROUND

The University of Texas at Austin, TX **January 2021 – May 2026**
Ph.D. in Aerospace Engineering & Engineering Mechanics (GPA: 3.81 / 4.0)

Indian Institute of Technology Madras, India **August 2017 – December 2019**
Master of Science in Structural Engineering (GPA: 9.41 / 10)

National Institute of Technology Calicut, India **August 2013 – August 2017**
Bachelor of Science in Civil Engineering (GPA: 9.15 / 10)

FELLOWSHIPS, SCHOLARSHIPS, and AWARDS

- Best poster award at the Workshop on Scientific Machine Learning, UT Austin. **October 2024**
- Warren A. and Alice L. Meyer Endowed Scholarship in Engineering from the Cockrell School of Engineering, UT Austin. **June 2024**
- Travel Awards by the United States Association for Computational Mechanics (June 2024), Society for Industrial and Applied Mathematics Texas-Louisiana (November 2022).
- Scholarship by the American Society of Indian Engineers and Architects (ASIE). **November 2022**
- Best MS Thesis award, Indian Institute of Technology Madras. **August 2020**
- Best Major B. Tech project award, National Institute of Technology, Calicut, India. **August 2017**
- Summer research fellowship, Department of mathematics, IISc, Indian Academy of Sciences. **July 2015**

JOURNAL PUBLICATIONS

- C. G. Krishnanunni., Tan Bui-Thanh., Clint Dawson (2025). Topological derivative approach for deep neural network architecture adaptation. (Link), Manuscript accepted at SIAM Journal on Scientific Computing. Contributions: Method development, theoretical analysis, and numerical implementation.
- C. G. Krishnanunni., Jonathan Wittmer., Tan Bui-Thanh., Quoc P. Nguyen (2026). A New Look at the Ensemble Kalman Filter for Inverse Problems: Duality, Non-Asymptotic Analysis and Convergence Acceleration. (Link), Under review at Inverse Problems. Contributions: Method development, theoretical analysis, and numerical implementation.
- William Cole Nockolds., C. G. Krishnanunni., Tan Bui-Thanh., Xianzhu Tang (2025). LiLaN: a linear latent network as the solution operator for real-time solutions to stiff and non-stiff nonlinear ordinary differential equations. *Machine Learning for Computational Science and Engineering* (Link) Contributions: Participation in method development and conducting the entire theoretical analysis.
- C. G. Krishnanunni., Tan Bui-Thanh (2025). An adaptive and stability-promoting layerwise training approach for sparse deep neural network architecture. *Computer Methods in Applied Mechanics and Engineering* (Link) Contributions: Method development, theoretical analysis, and numerical implementation.

- Jonathan Wittmer., C. G. Krishnanunni., Hai Van Nguyen., Tan Bui-Thanh (2023). On Unifying Randomized Methods for Inverse Problems. *Inverse Problems*. (Link) Contributions: Participation in theoretical analysis and numerical implementation.
- KS Suraj., PM Anilkumar., C. G. Krishnanunni., BN Rao (2023). Uncertainty quantification of bistable variable stiffness laminate using machine learning assisted perturbation approach. *Composite Structures* (accepted) (Link) Contributions: Participation in method development and theoretical analysis.
- Albert Orwa Akuno., L. Leticia Ramirez-Ramirez., Chahak Mehta., C. G. Krishnanunni., Tan Bui-Thanh., Jose Arturo Montoya (2022). Multi-patch epidemic models with partial mobility, residency, and demography. *Chaos, Solitons, & Fractals*. Contributions: Participation in theoretical analysis (Link)
- C. G. Krishnanunni., B. N. Rao., (2021). Indirect health monitoring of bridges using Tikhonov regularization scheme and signal averaging technique. *Structural Control and Health Monitoring*. (Link) Contributions: Method development, conducting the entire theoretical analysis, and numerical implementation

More publications details are available HERE

RECENT INVITED TALKS

- Topological derivative approach for deep neural network architecture adaptation. *Computational Algebra Seminar*, Department of Mathematics, UT Austin, May 2, 2025.
- Layerwise sparsifying training and sequential learning strategy for neural architecture adaptation. *U. S. National Congress on Computational Mechanics*, New Mexico, July 23-27, 2023.
- A two-stage strategy for neural architecture adaptation. 5^{th} *Annual meeting of the SIAM Texas-Louisiana Section on Uncertainty Quantification*, Houston, November 4-6, 2022.

More invited talks are available HERE

RECENT RESEARCH EXPERIENCES

Solving forward and inverse problems in Plasma fusion research
Collaborator: Dr. Tan Bui-Thanh (UT Austin) & Dr. Xianzhu Tang (Los Alamos National Laboratory)

- Research aimed at i) adaptive strategies for learning a surrogate for the Collisional-Radiative (CR) model; and ii) solving an inverse problem for runaway electron mitigation during a plasma disruption event using evolutionary algorithms.

Transformer-powered generative model for solving inverse problems via joint modeling with forward process
Collaborator: Dr. Kowshik Thopalli, Dr. Yamen Mubarka, Dr. Vivek Narayanaswamy, Dr. Jayaraman J. Thiagarajan (Lawrence Livermore National Laboratory, USA)

- Designed a transformer architecture based generative model that transports samples from a prior distribution to samples from posterior parameter distribution conditioned on an input measurement.

Developing efficient algorithms for neural architecture adaptation
Collaborator: Dr. Tan Bui-Thanh (UT Austin, USA)

- Research in mathematical optimization and machine learning aimed at developing a mathematically principled way for automatically determining neural network architecture for a given data-set.

A new look at the Ensemble Kalman filter via duality
Collaborator: Dr. Tan Bui-Thanh (UT Austin, USA)

- Research aimed at the analysis of Ensemble Kalman filter for inverse problems in order to get insights into new convergence improvement strategies.

Mathematical epidemiology project
Collaborator: Dr. Tan Bui-Thanh (UT Austin, USA) & Leticia Ramirez-Ramirez (CIMAT, Mexico)

- Research aimed at developing an epidemic model that takes into account the effects of human mobility on the evolution of disease dynamics in a multi-population environment.

LEADERSHIP/MENTORSHIP ROLES

- **Moncrief Summer Internship mentor**:
  - ∗ Mentored Jennifer Zheng on a project titled *Physics informed deep-learning approach enhanced by POD for forecasting solutions to time-dependent PDE.*
  - ∗ Mentored Giancarlo Villatoro on a project titled From an Elementary Proof of Error Representation for Hermite Quadrature to a Rediscovery of Legendre Polynomials and Rodrigues Formula
- **SIAM-UT Mentorship program:** Worked with Venkata Hasith Vattikuti on a project focused on the use of reinforcement learning for solving a combinatorial optimization problem (nonlinear dimension reduction).
- **Roles in Conferences:** Co-organized mini-symposiums at USNCCM17, 5th SIAM TX-LA Annual Meeting, SIAM Conference on Mathematics of Data Science, Member of the local organizing committee for 8th SIAM TX-LA Annual Meeting held at UT Austin.

PROFESSIONAL EXPERIENCE

Graduate Teaching/Research Assistant — **January 2021 - Present**
Oden Institute of Computational Engineering & Sciences, UT Austin — *Austin, TX*

- Research assistant to Prof. Tan Bui-Thanh, Institute of Computational Engineering and Sciences.
- Teaching assistant for courses: Analytical methods, Mathematical methods in Science and Engineering.

Computing graduate student intern — **June 2024 - August 2024**
Lawrence Livermore National Laboratory — *Livermore, California*

- Research intern at the Machine Intelligence Group, LLNL.

Graduate Teaching/Research Assistant — **August 2017 - December 2020**
Indian Institute of Technology Madras — *Madras, India*

- Research assistant to Prof. B. N. Rao, Structural Engineering department, IIT Madras.
- Teaching assistant for courses: Structural optimization and Finite element analysis.

RESEARCH FUNDING AGENCIES

My research has been funded by the following agencies:
United States Department of Energy: Worked on the grant titled "DeepFusion Accelerator for Fusion Energy Sciences in Disruption Mitigation". UT Austin PI: Tan Bui-Thanh.
United States National Science Foundation: Worked on the grant titled "Toward a Rigorous and Reliable Scientific Deep Learning Framework for Forward, Inverse, and UQ Problems". UT Austin PI: Tan Bui-Thanh.
ConTex: Worked on the grant titled "Machine-Learning Assisted Real-Time Simulations and Uncertainty Quantification for Infectious Disease Outbreaks". UT Austin PI: Tan Bui-Thanh.

JOURNAL ROLES

Peer Reviewer (Journals): *Applied Mathematical Modelling,* Elsevier, *Sound and Vibration,* Elsevier, *Applied Ocean Research*, Elsevier.
Reviewer (Conferences): *SIAM MDS24, USNCCM17, 8th SIAM TX-LA Annual Meeting.*

SKILLS

Software: MATLAB®, LaTeX®, ANSYS ®, FEniCS ®

Programming Languages: C++, Java, Python

ML Library: TensorFlow, PyTorch, JAX
HPC & Computing: GPU-accelerated model training, high-performance computing on TACC (Texas Advanced Computing Center) clusters.

REFERENCES

- **Tan Bui-Thanh**
  Professor,
  Leader of Pho-Ices group
  Department of Aerospace Engineering and Engineering Mechanics
  The Oden Institute for Computational Engineering and Sciences
  The University of Texas at Austin
  Austin, USA
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- **Xianzhu Tang**
  Physicist
  Los Alamos National Laboratory
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- **Kentaro Yaji**
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